

Bota Andrei-Daniel

+40 770-434-062 | bota.co.andrei@student.utcluj.ro | github.com/BotaAndreiD | linkedin.com/in/andrei-bota

Applied Electronics graduate (Technical University of Cluj-Napoca, 2026) drawn to analog/mixed-signal circuit design and PCB layout — from schematic capture to a fabricated, standards-compliant board — with supporting experience in embedded firmware and FPGA development. Seeking an analog design or PCB layout engineering role.

EDUCATION

Technical University of Cluj-Napoca

Oct 2022 – Jul 2026

B.Sc. in Applied Electronics — Faculty of Electronics, Telecommunications and Information Technology · Cluj-Napoca, Romania

- Cumulative GPA: 8.24/10
- Bachelor's Thesis — "CardioMed: Portable ECG Acquisition and Monitoring System" — graded 10/10

Solomon Haliță Theoretical High School

Sep 2018 – Jul 2022

Mathematics and Computer Science specialization · Sângeorz-Băi, Romania

HONORS & AWARDS

- Special Honor for CardioMed, SSET 2026 Student Symposium — awarded by Analog Devices (May 2026)
- 2nd Honorable Mention for CardioMed, SSET 2026 Student Symposium (May 2026)
- Merit Scholarship for Academic Performance, UTCN (2024–2026)

PROFESSIONAL EXPERIENCE

Analog Devices — Research & Development Intern

Jul 2025 – Aug 2025

Cluj-Napoca, Romania

- Collaborated on the development of an integrated hardware-software system, applying electronics and embedded programming knowledge to real-world industrial challenges.
- Conducted technical analysis, testing, and documentation within a professional R&D environment.

Popeye Fast-Food — Service Staff

Jul 2019 – Aug 2019

Bistrița-Năsăud, Romania

- Managed customer orders and interactions in a fast-paced environment, building early experience in communication, teamwork, and quick problem-solving under pressure.

TECHNICAL PROJECTS

CardioMed — Portable ECG Acquisition & Monitoring System

Oct 2025 – Jul 2026

Bachelor's Thesis

- Designed a discrete analog front-end — input protection, Right Leg Drive loop, instrumentation amplifier, and cascaded Bessel / Twin-T notch / Butterworth active filters — as a low-cost alternative to integrated AFE chips, targeting IEC 60601-2-25 compliance.
- Achieved a simulated CMRR of 123–137 dB and 10.76 μ V input-referred noise, roughly 3x better than the standard's 30 μ V threshold.
- Designed a compact 4-layer, 82×46 mm PCB in KiCad with analog/digital domain separation and differential trace routing; digitized signal via a 24-bit ADC and transmitted it wirelessly over Bluetooth Low Energy (ESP32-C3).
- Built a companion C#/WPF desktop application performing real-time PQRST detection (Pan–Tompkins-based algorithm), computing 7 clinical parameters, and auto-generating PDF reports.
- Validated detection algorithms against public PhysioNet databases (MIT-BIH, QT Database), achieving 91.5% sensitivity for R-wave detection; delivered a fully simulated, production-ready design at ~130 RON BOM cost per unit at 1,000-unit volume.
- Tools: LTspice, KiCad, C# / WPF, Embedded C, Python, Bluetooth Low Energy

FPGA-Based Hardware-Software Module (AD5592R)

Jul 2025 – Aug 2025

Analog Devices Summer School

- Delivered a complete hardware-software system integrating an ADC and accelerometer on a Xilinx FPGA platform, covering the full stack from board-level hardware to application software.
- Designed the schematic and PCB in KiCad, implemented custom HDL logic in Verilog for the Cora Z7 FPGA, and developed an Embedded Linux SPI driver for the AD5592R.
- Built a Python application for real-time data acquisition and processing, translating live accelerometer readings into mouse-control input.
- Tools: KiCad, Vivado, Verilog, Embedded Linux, SPI, Python, GitLab CI/CD

ADDITIONAL TRAINING & CERTIFICATIONS

- Machine Learning Course (UTCN) — fundamentals of ML, data preprocessing, and classification/regression with Python and scikit-learn.
- Analog Devices Workshops — analog electronics, control systems, and embedded development on Linux and bare-metal.
- SMART PRACTICE Project Certificate — professional training program aligning academic knowledge with labor-market requirements through practical internships.

TECHNICAL SKILLS & LANGUAGES

Programming: C, C++, C# (WPF), Python, Java, Verilog/HDL

Hardware Design & Tools: KiCad, OrCAD, Proteus, Vivado (Xilinx), LTspice, MATLAB, Keil

Platforms: FPGA (Cora Z7, Artix-7), Embedded Linux, ESP32 / Bluetooth Low Energy, ADALM2000, Analog Discovery, Scopy

Other: GitLab CI/CD, GitHub

Languages: English (B2), Romanian (Native), German (A1)